Personally Identifying Information (PII) redactor for web-scraped Online Job Advertisements

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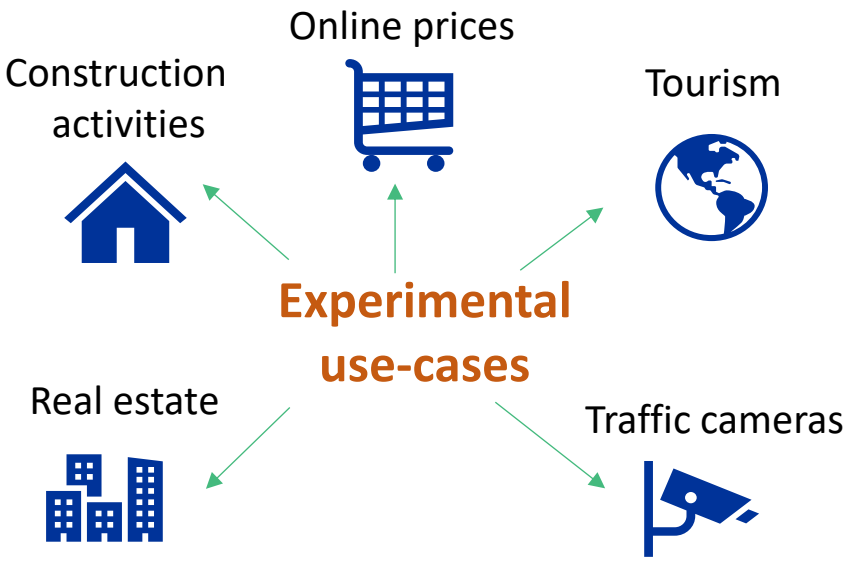
European Commission

Eurostat’s Web Intelligence Hub

Eurostat started investigating the use of web scraping for European statistics in 2017. In 2020 Eurostat established the Web Intelligence Hub (WIH). The WIH provides the infrastructure and expertise to support the use of web content for European Statistics.

Over the years, the WIH has explored several use cases. Two use cases have reached a higher level of maturity: online job advertisements (OJAs) and online business enterprise characteristics (OBEC).



Most mature use-cases

Online job advertisements OJA

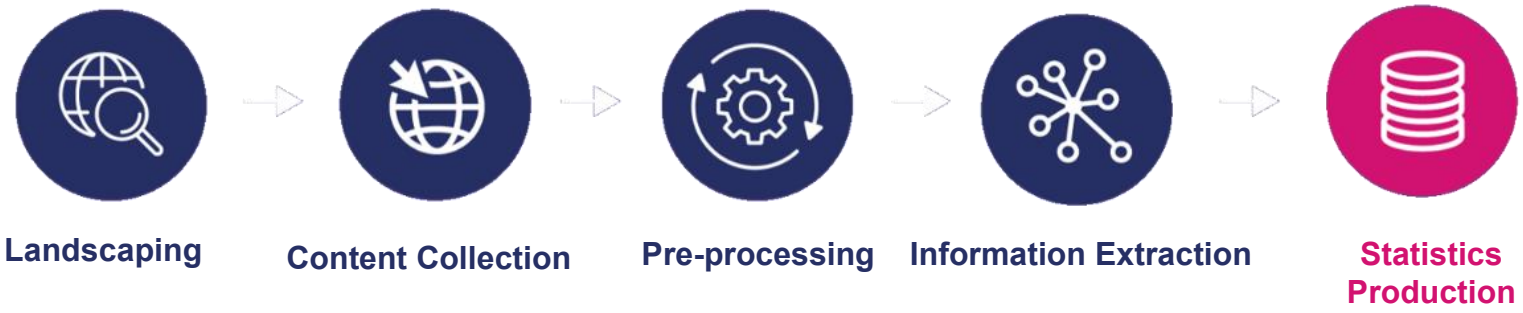
Enterprise characteristics OBEC

Web Intelligence Hub - Online Job Advertisements

Online Job Advertisements (OJA) from job portals have been identified as a promising data source to track the labour market demand for occupations and skills.

Since 2020, Eurostat has established and maintained a pipeline for the collection and processing of online job advertisements. Eurostat currently publishes two experimental statistics using online job advertisements data:

- Job vacancy rate by occupation and region
- Labour market demand for ICT specialists in OJAs



Details on the OJA data collection are available on Eurostat's CROS portal.

OJA data collection and personal data

As per the GDPR, personal data refers to any information relating to an identified or identifiable natural person (‘data subject’).

We call Personally Identifiable Information (PII) any data that can be used to identify a specific individual, either directly or indirectly, and includes things like names, addresses, social security numbers, biometric identifiers, email addresses, and many other data

Eurostat’s OJA data collection does lead to the collection of several types of PII :

- Names of people (first names, family names)
- Social profiles URLs (LinkedIn, Instagram, Twitter, X, TikTok, Facebook)
- Phone numbers
- Email addresses

General Data Protection Regulation (GDPR)

The GDPR regulates the processing of personal data within the EU.

Article 89 establishes derogations from several of the general rights of data subjects where data are processed for scientific or historical research purposes or statistical purposes. However:

- The processing must still be lawful (Article 6)
- The principle of data minimization still applies (Article 5)
- Appropriate safeguards (such as pseudonymization) should be put in place to protect the rights of data subjects (Article 89)

The fact that certain personal data is public does not strip it of its protections!

General Data Protection Regulation (GDPR)

- The processing must still be lawful (Article 6)

Lawful under Article 6(1)(e) of the GDPR, which permits data processing when it is necessary for the performance of a task carried out in the public interest, as specified by EU or national law. In this case, regulation (EC) No 223/2009 on European Statistics. Provides the legal basis for processing.

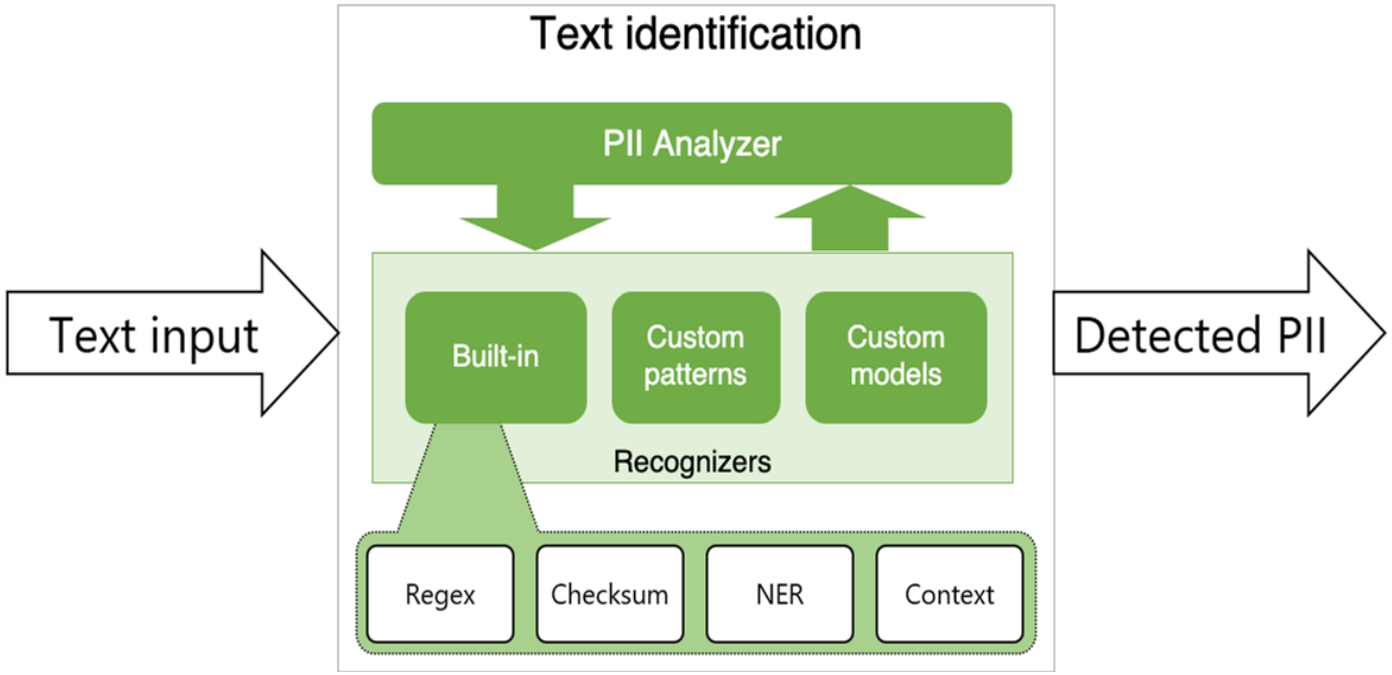
- The principle of data minimization still applies (Article 5)
- Appropriate safeguards (such as pseudonymization) should be put in place to protect the rights of data subjects (Article 89)

As we do not really need the personal data for our purposes, the approach chosen is to redact PII immediately upon collection.

Microsoft PRESIDIO

Presidio (Origin from Latin praesidium ‘protection, garrison’) is an open-source framework for identifying and anonymizing in text and images.

It provides a flexible and customizable framework that includes both pre-defined PII recognisers and allows custom or external recognisers to be plugged in.



Presidio Detection Flow

Regex
pattern recognition

NER(ML)*
leveraging natural language to detect entities

Checksum
validate patterns (if applicable)

Context Words
increase the detection confidence

Anonymization
multiple anonymization techniques

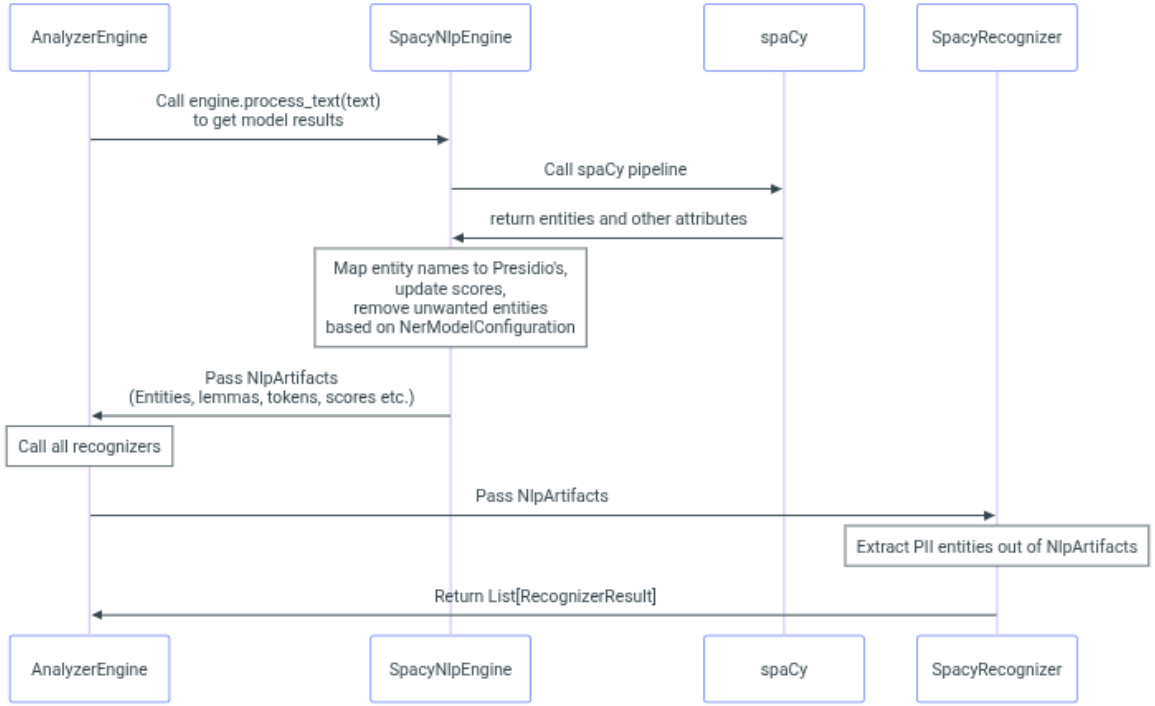
*NER – Named Entity Recognition

PRESIDIO analysers for OJA

Presidio Analyzers can use pre-trained or custom models coming from spaCy or Stanza

https://spacy.io/usage/models >>> Natural Language Processing in Python

https://stanfordnlp.github.io/stanza/index.html >>> Stanza is a collection of accurate and efficient tools for the linguistic analysis of many human languages



Both spaCy/Stanza frameworks includes multi-language models and also language-specific models for a large list of languages.

PRESIDIO Recognizers for OJA data

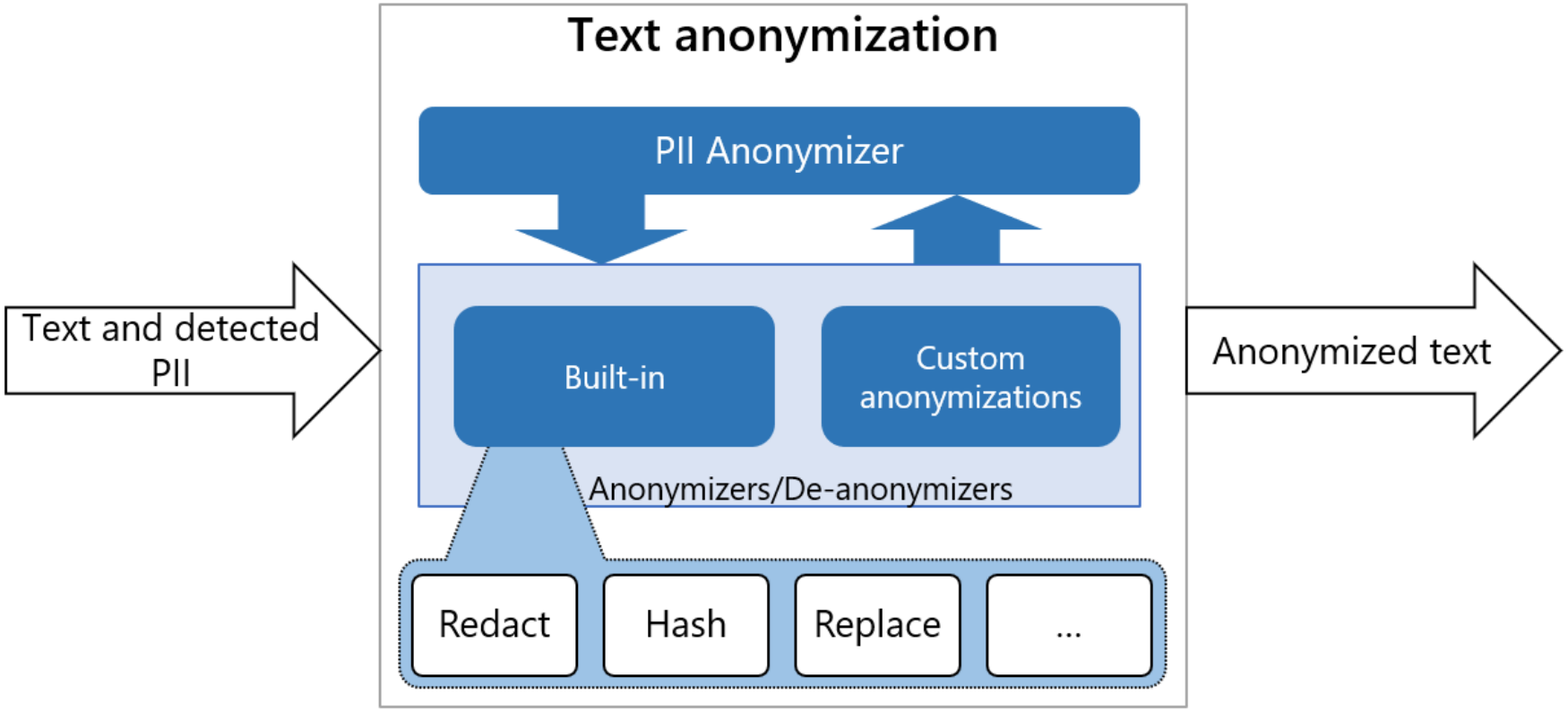
RegEx based Recognizer

- Email addresses
- Social media URLs
- Phone numbers. This recognisers expects the phone number to follow some predefined format – a simple “sequence of numbers” is not recognised as a phone number

NPL-model based Recognizers

- Phone numbers (SpaCy/Stanza)
- Names: ***XLM-RoBERTa:51la5/roberta-large-NER*** model from Hugging Face. XLM-RoBERTa is a multilingual model trained on 100 different languages

Anonymizer


Text anonymization
PII Anonymizer
Text and detected PII
Built-in
Custom anonymizations
Anonymizers/De-anonymizers
Redact
Hash
Replace
...
Anonymized text

Anonymization tags

Hi, my name is David and my number is 212 55 1234

Hi, my name is <NAME> and my number is <PHONE_NUMBER>

Default tags <NAME>, <URL>, <PHONE_NUMBER> and <EMAIL_ADDRESS> break the HTML code

- ***NAME***: Names of people (first names, family names)
- ***URL***: Social profiles URLs
- ***PHONE_NUMBER***: Phone numbers
- ***EMAIL_ADDRESS***: Email addresses

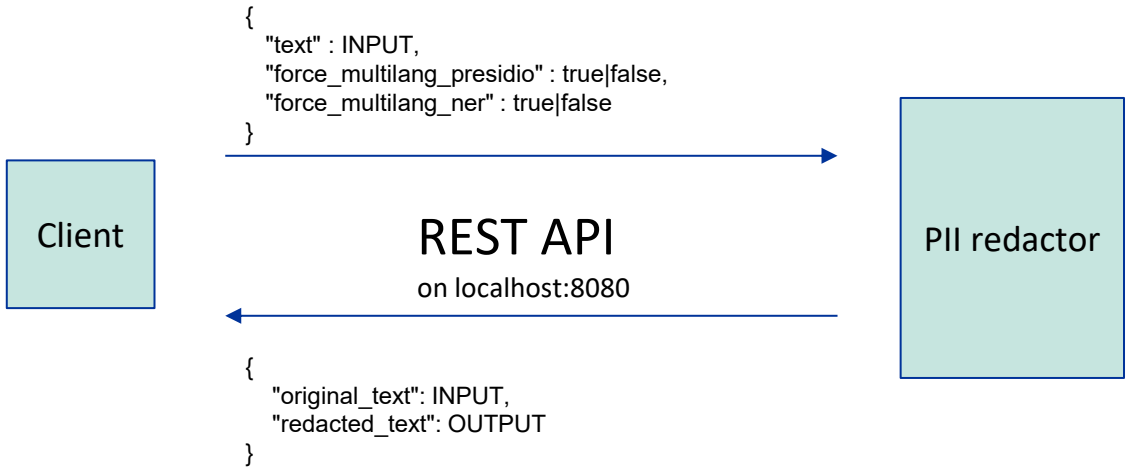
Ongoing testing and next steps

- Checking the accuracy of the redaction, using a (small) sample of both real OJAs and synthetic OJAs generated by AI.
- Checking that the information redaction does not have an important effect on the classification of OJAs.
- Checking the costs (monetary + processing time) of applying the PII redactor.

Following completion of these tests, the PII redactor will be applied to newly collected data and to data made available to researchers (e.g. the “NLP sample” dataset, which contains the full text of job advertisements)

Deployment

The PII redactor will be packaged as a Docker image and run as a service in a Docker container within the Web Intelligence Hub. It will handle input in plain text, JSON and HTML.



Reuse (either within the WIH or by deploying the Docker image elsewhere) is possible, **but** some of the choices made in terms of which entities to recognize / redact were based on the OJA use case…

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Thank you

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